Fuzzy Logic and Fuzzy Set Theory

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What is Fuzzy Logic?

- An approach to computing based on degrees of truth rather than the usual true/false (1 or 0).
- Proposed by Lotfi Zadeh in 1965.
- Used to handle the concept of partial truth.

Introduction to Classical Sets

- Defined by crisp boundaries.
- Elements either belong or do not belong to a set.
- Operations: Union, Intersection, Complement, Difference.

Introduction to Fuzzy Sets

- Elements have degrees of membership [0,1].
- Represent uncertainty and vagueness.
- Defined using membership functions.

Operations: Classic vs Fuzzy Sets

Classic Sets:

- Union: $A \cup B$
- Intersection: $A \cap B$
- Complement: A^c
- Difference: $A - B$

Fuzzy Sets:

- Union:
  $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$
- Intersection:
  $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$
- Complement:
  $\mu_{A^c}(x) = 1 - \mu_A(x)$
- Difference: $\mu_{A-B}(x) =$
  $\min(\mu_A(x), 1 - \mu_B(x))$

Properties of Sets (Mathematical Explanation)

Commutativity

- Classical: $A \cup B = B \cup A$, $A \cap B = B \cap A$
- Fuzzy: $\max(\mu_A(x), \mu_B(x)) = \max(\mu_B(x), \mu_A(x))$

Associativity

- Classical: $(A \cup B) \cup C = A \cup (B \cup C)$
- Fuzzy:
  $\max(\mu_A(x), \max(\mu_B(x), \mu_C(x))) = \max(\max(\mu_A(x), \mu_B(x)), \mu_C(x))$

Properties of Sets (Continued)

Distributivity

- Classical: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- Fuzzy: $\min(\mu_A(x), \max(\mu_B(x), \mu_C(x))) = \max(\min(\mu_A(x), \mu_B(x)), \min(\mu_A(x), \mu_C(x)))$

Idempotency

- Classical: $A \cup A = A,\ A \cap A = A$
- Fuzzy: $\max(\mu_A(x), \mu_A(x)) = \mu_A(x),\ \min(\mu_A(x), \mu_A(x)) = \mu_A(x)$

More Properties of Sets

Identity

- Classical: $A \cup \emptyset = A$, $A \cap U = A$
- Fuzzy: $\max(\mu_A(x), 0) = \mu_A(x)$, $\min(\mu_A(x), 1) = \mu_A(x)$

Transitivity (mainly for relations)

- Classical: If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$
- Fuzzy: $\mu_R(a, c) \geq \min(\mu_R(a, b), \mu_R(b, c))$

De Morgan's Laws

- Classical: $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$
- Fuzzy: $1 - \max(\mu_A(x), \mu_B(x)) = \min(1 - \mu_A(x), 1 - \mu_B(x))$

Crisp Relations

- Cartesian Product: $A \times B$
- Universal Relation: All pairs
- Identity Relation: (a, a) for all $a \in A$
- Cardinality: Number of pairs in relation

Operations on Crisp Relations

- Union: $R_1 \cup R_2 = \{(a, b) \mid (a, b) \in R_1 \text{ or } (a, b) \in R_2\}$
- Intersection: $R_1 \cap R_2 = \{(a, b) \mid (a, b) \in R_1 \text{ and } (a, b) \in R_2\}$
- Complement: $\overline{R} = \{(a, b) \mid (a, b) \notin R\}$
- Composition: If $R \subseteq A \times B$ and $S \subseteq B \times C$, then:

$$R \circ S = \{(a, c) \mid \exists b \in B, (a, b) \in R \text{ and } (b, c) \in S\}$$

Fuzzy Relations



- Extension of crisp relations with membership values:

$$R = \{(x, y, \mu_R(x, y))\}$$

- Example:

$$R = \{(a, b, 0.8), (a, c, 0.3), (b, c, 1.0)\}$$

- Represent vague associations like:
  - "a is strongly related to b"
  - "a is weakly related to c"
- Used in similarity, preferences, and decision-making problems.

Operations of Fuzzy Relations

- Max-Min Composition:

$$\mu_{R \circ S}(x, z) = \max_y \min(\mu_R(x, y), \mu_S(y, z))$$

- Max-Product Composition:

$$\mu_{R \circ S}(x, z) = \max_y (\mu_R(x, y) \cdot \mu_S(y, z))$$

- **Example:**
  - Let $R = \begin{bmatrix} 0.2 & 0.8 \\ 1.0 & 0.5 \end{bmatrix}$, $S = \begin{bmatrix} 0.9 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$
  - Max-Min:
    $\mu_{R \circ S}(1, 1) = \max(\min(0.2, 0.9), \min(0.8, 0.3)) = \max(0.2, 0.3) = 0.3$
  - Max-Product:
    $\mu_{R \circ S}(1, 1) = \max(0.2 \cdot 0.9, 0.8 \cdot 0.3) = \max(0.18, 0.24) = 0.24$

Equivalence Tolerance Relations

- Equivalence: Reflexive, symmetric, transitive
- Tolerance: Reflexive, symmetric

Cosine Amplitude

Definition: Measures the **angle similarity** between two fuzzy sets A and B.

$$S_{cos}(A, B) = \frac{\sum_x \mu_A(x) \cdot \mu_B(x)}{\sqrt{\sum_x \mu_A(x)^2} \cdot \sqrt{\sum_x \mu_B(x)^2}}$$

Interpretation: Values close to 1 indicate high similarity in direction (shape).

Max-Min Similarity



Max-Min Similarity

Definition: Based on the **maximum of the minimum** membership values.

$$S_{max\text{-}min}(A, B) = \max_x \left(\min\left(\mu_A(x), \mu_B(x)\right)\right)$$

Interpretation: Captures the strongest point-wise agreement.

Fuzzification

- **Crisp Input:** A precise, numerical input value from the real world (e.g., temperature = 37°C).
- **Fuzzification:** Process of mapping crisp input values to fuzzy sets using membership functions.
- **Membership Value:** A number in [0,1] that indicates the degree to which the input belongs to a fuzzy set.
- **Example:** Temperature = 37°C may belong to:
  - "Cold" with $\mu = 0.0$
  - "Warm" with $\mu = 0.7$
  - "Hot" with $\mu = 0.3$

Understanding Fuzzy Reasoning Techniques

Intuition:

- Based on human-like perception and gut feeling.
- *Example:* "This road seems slippery, I should slow down," even without concrete evidence.

Inference:

- Drawing conclusions from fuzzy rules.
- *Example:* If temperature is high and humidity is low, then fire risk is high.

Rank Ordering:

- Sorting based on degrees of membership or preference.
- *Example:* Prioritizing patients in triage: Critical (0.9), Serious (0.7), Stable (0.4).

Inductive Reasoning:

- Learning general fuzzy rules from examples or observations.
- *Example:* Observing that most tall people reach high shelves $\rightarrow$ Rule: "Tall people can reach high places."

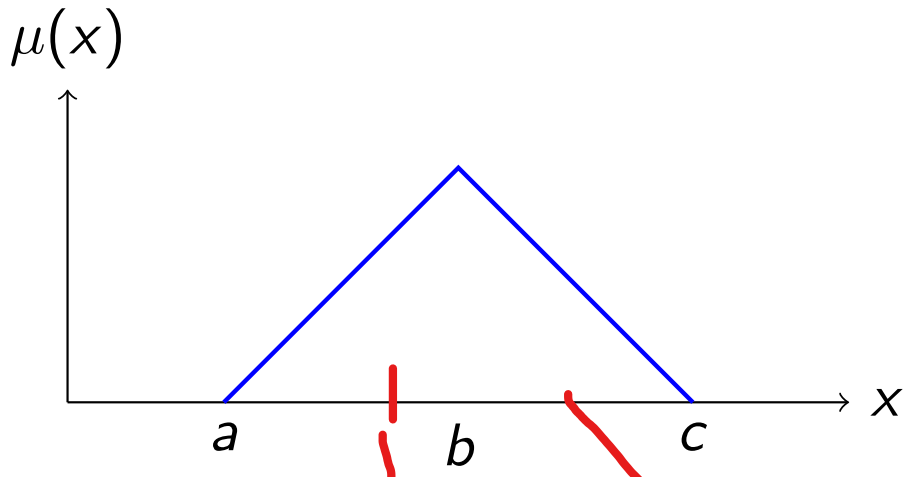
Fuzzy Membership Function

- Core: Elements with membership $= 1$
- Support: Membership > 0
- Boundary: Membership between 0 and 1
- Normal Fuzzy Set: Max membership $= 1$
- Convex Fuzzy Set: All points between two members also included
- Crossover Points: Membership $= 0.5$

Triangular Membership Function

Definition

A triangular membership function has a peak and linearly decreases on both sides.



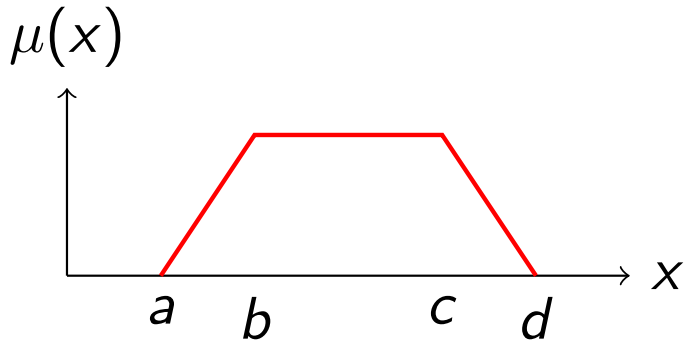
Equation

$$\mu(x) = \begin{cases} 0 & x \leq a \text{ or } x \geq c \\ \frac{x-a}{b-a} & a < x \leq b \\ \frac{c-x}{c-b} & b < x < c \end{cases}$$

Trapezoidal Membership Function

Definition

A trapezoidal function has a flat top between b and c and linearly increases/decreases.



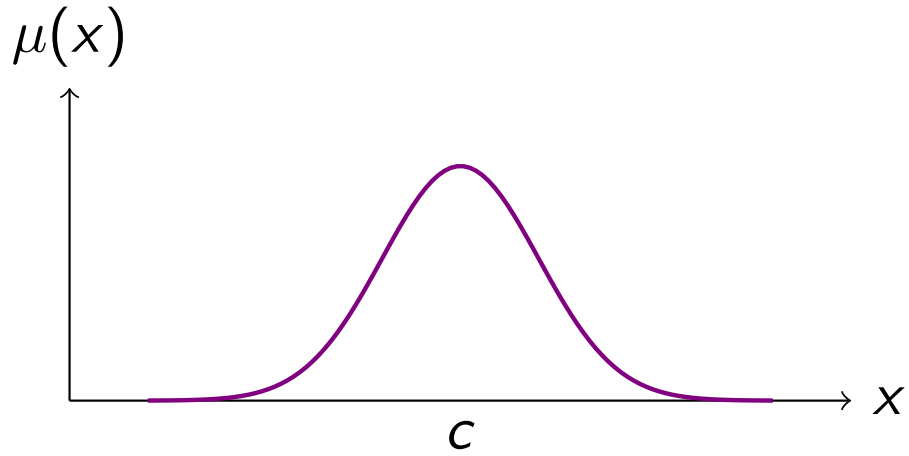
Equation

$$\mu(x) = \begin{cases} 0 & x \leq a \text{ or } x \geq d \\ \frac{x-a}{b-a} & a < x \leq b \\ 1 & b < x \leq c \\ \frac{d-x}{d-c} & c < x < d \end{cases}$$

Gaussian Membership Function

Definition

Smooth bell-shaped curve, useful for gradual transitions.



Equation

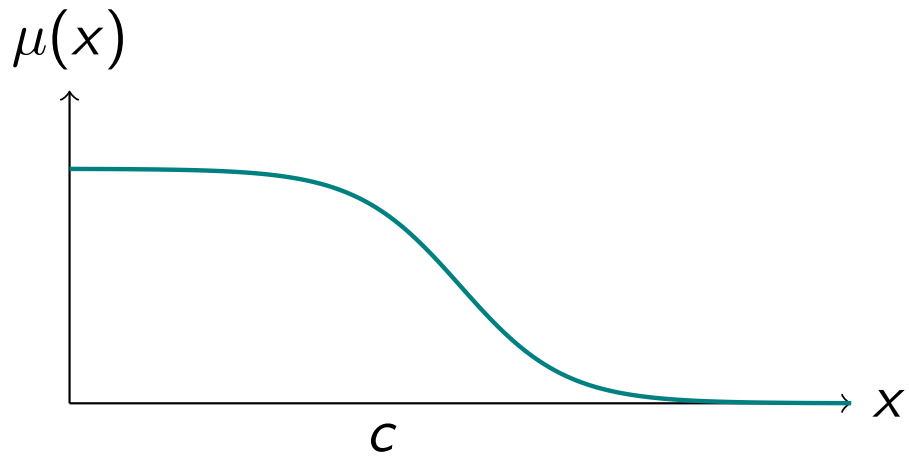
$$\mu(x) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right)$$

Where c is the center(mean) and σ controls the width(variance).

Sigmoidal Membership Function

Definition

S-shaped curve, typically used for open-ended fuzzy sets.



Equation

$$\mu(x) = \frac{1}{1 + \exp[-a(x - c)]}$$

Where c is the inflection point and a determines the slope.

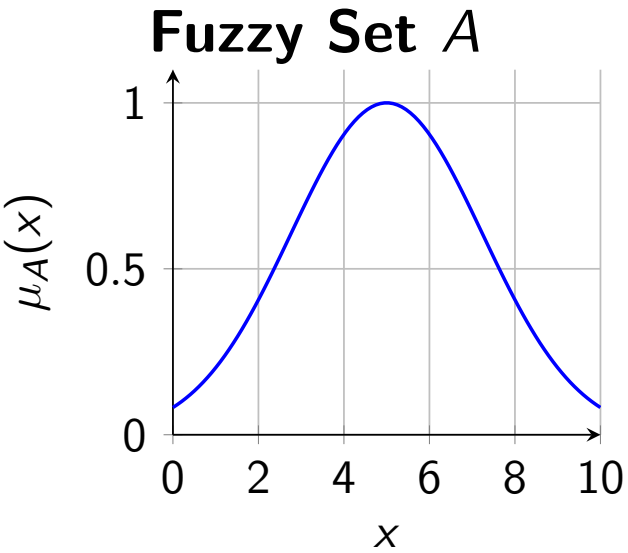
Summary of Membership Functions

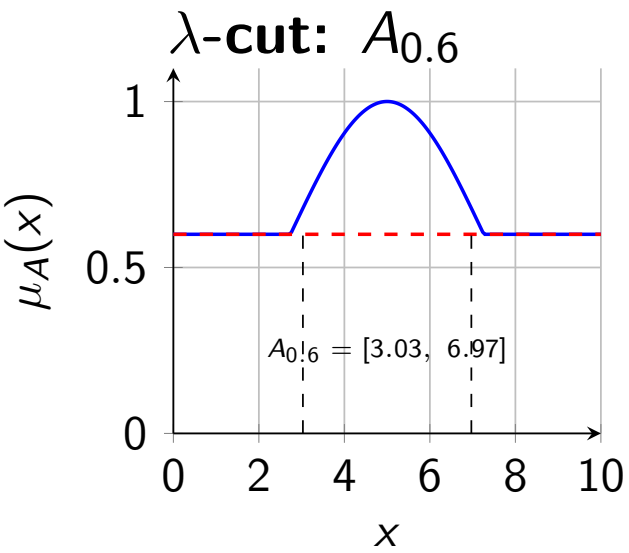
Type	Equation	Parameters
Triangular	Piecewise linear	a, b, c
Trapezoidal	Piecewise linear with plateau	a, b, c, d
Gaussian	$e^{\left(-\frac{(x-c)^2}{2\sigma^2}\right)}$	c, σ
Sigmoidal	$\frac{1}{1+e^{[-a(x-c)]}}$	a, c

λ-cut and Defuzzification

- λ-cut: Converts fuzzy set to crisp by threshold λ
- Defuzzification: Converting fuzzy output to crisp value
  - Centroid Method
  - Bisector Method
  - Mean of Maximum (MoM)

Graphical View of λ-cut (Before and After)

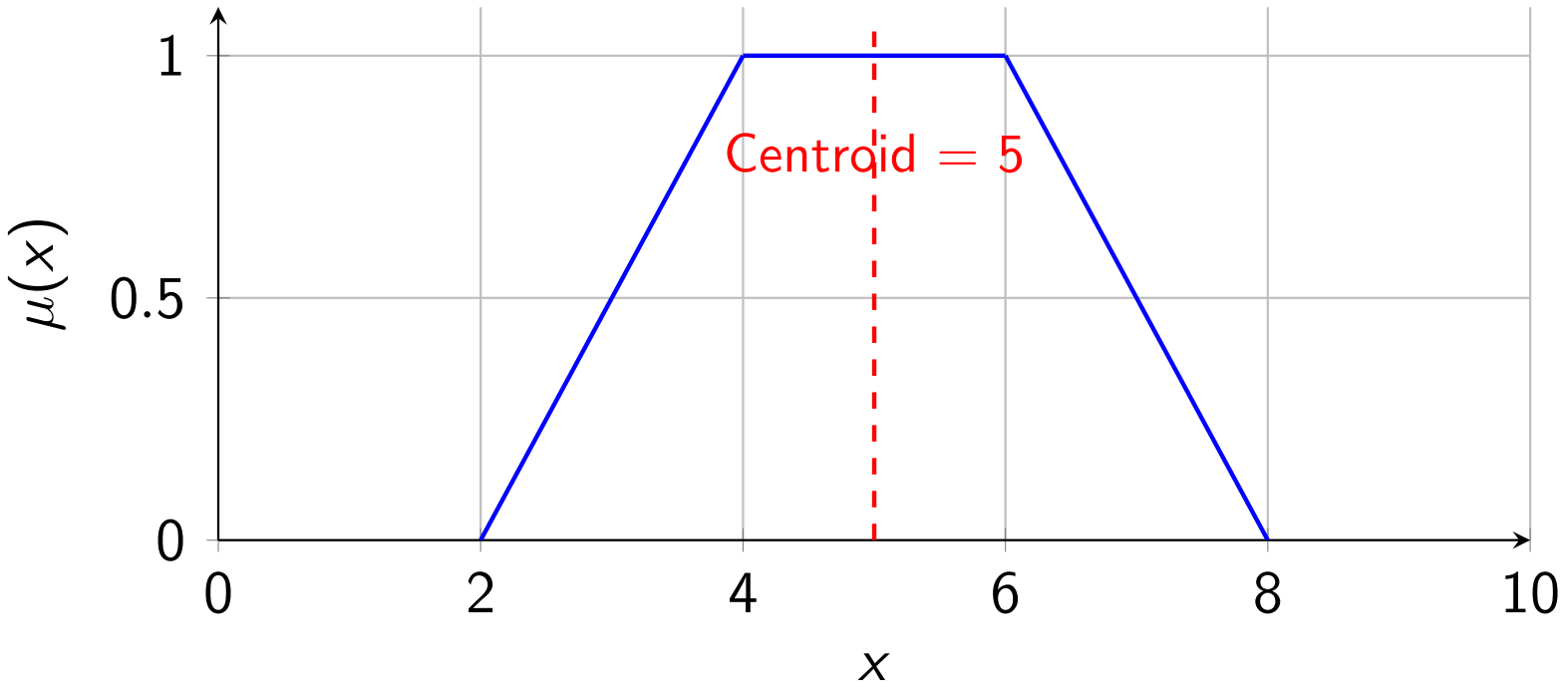




Defuzzification: Centroid Method

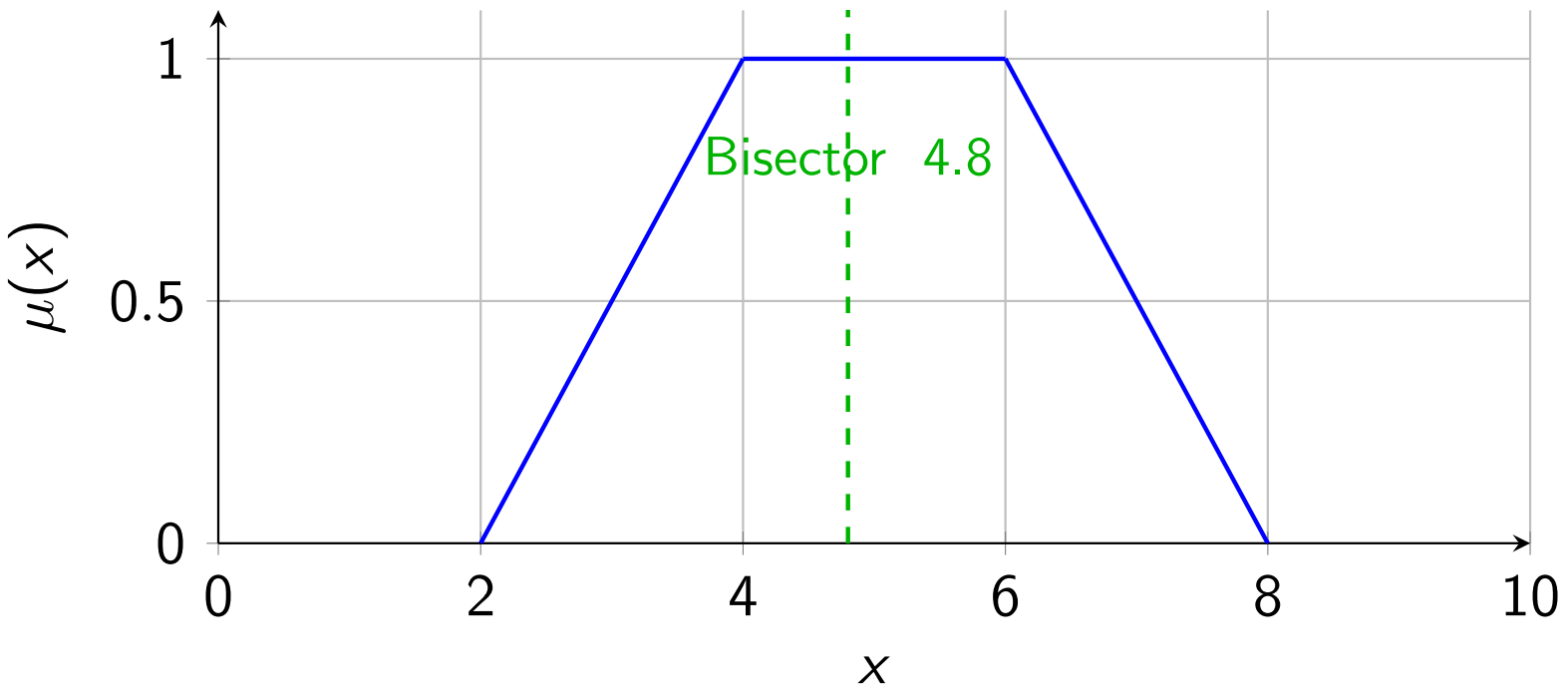
- Calculates the **center of gravity** (center of area) of the fuzzy set.
- Most widely used and accurate method.
- Formula:

$$x^* = \frac{\int x \cdot \mu(x)\, dx}{\int \mu(x)\, dx}$$



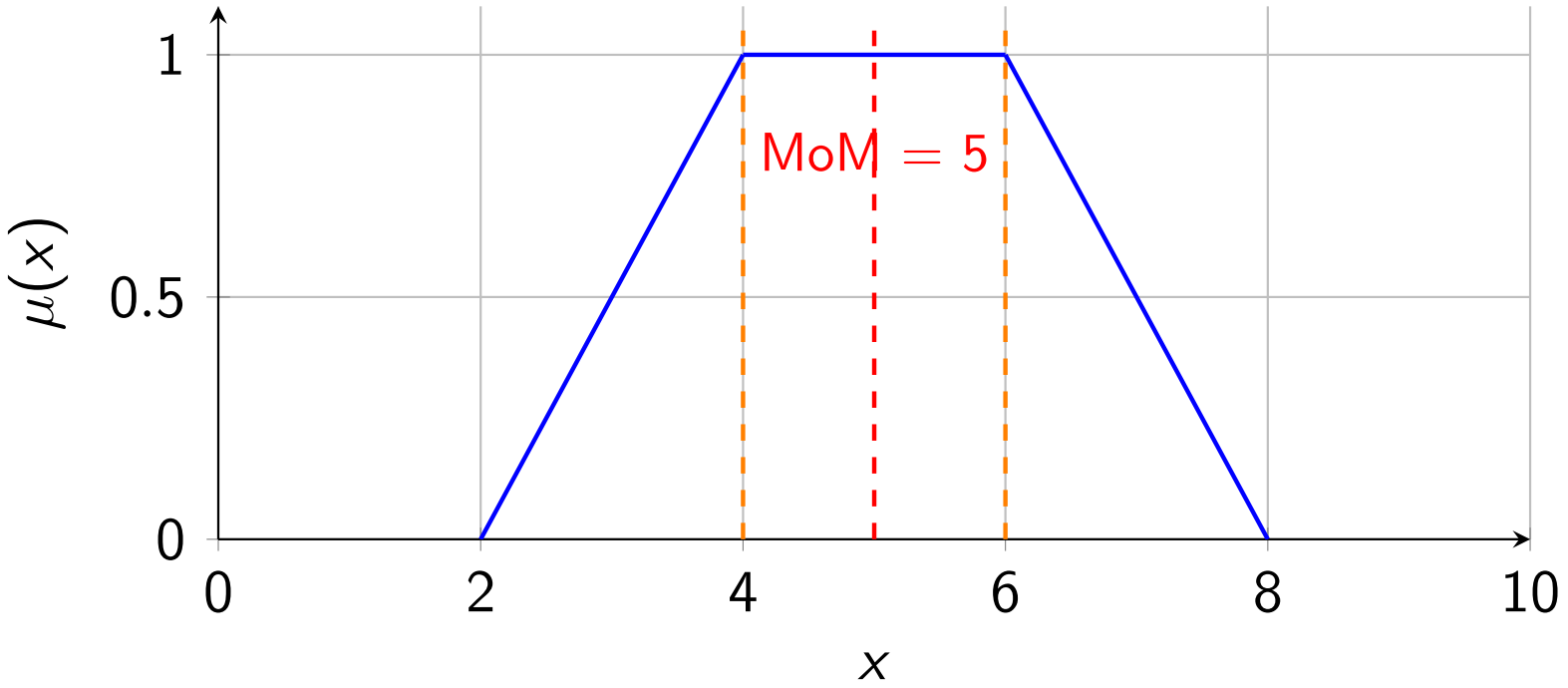
Defuzzification: Bisector Method

- Finds a vertical line that divides the fuzzy area into two equal halves.
- Not always equal to the centroid.



Defuzzification: Mean of Maximum (MoM)

- Averages all values of x where $\mu(x)$ is maximum.
- Easy to compute.



Fuzzy Logic: Single-Temperature Controller

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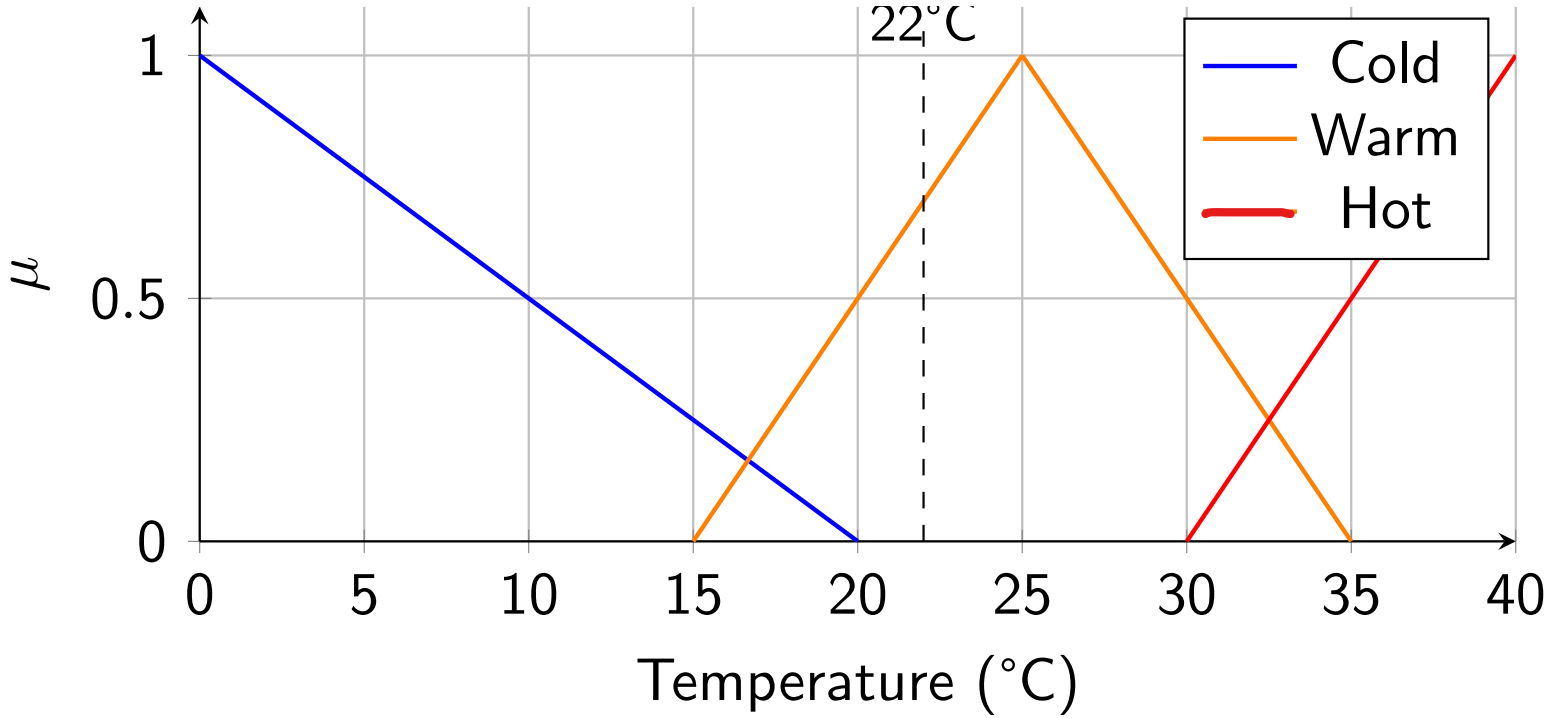
Problem Statement



- Goal: Control a heater based on room temperature using fuzzy logic.
- Input: Room temperature (°C)
- Output: Heater power level (0% to 100%)
- Temperature range: 0°C to 40°C
- Linguistic Variables:
  - Temperature: **Cold, Warm, Hot**
  - Power: **High, Medium, Low**

Fuzzification

- Input temperature is converted into fuzzy values.
- Example: Room temperature = 22°C



Fuzzy Membership Values at 22°C

- $\mu_{\text{Cold}}(22) = 0$ (outside range)
- $\mu_{\text{Warm}}(22) = \frac{22-15}{10} = 0.7$
- $\mu_{\text{Hot}}(22) = 0$ (outside range)

Here, (in the warm triangular)
x=22, a=15(start), b=25(middle), c=35(end)
Here, a<x≤b
so, x-a / b-a = 22-15 / 25-15

Conclusion: The temperature is **Warm** with degree 0.7

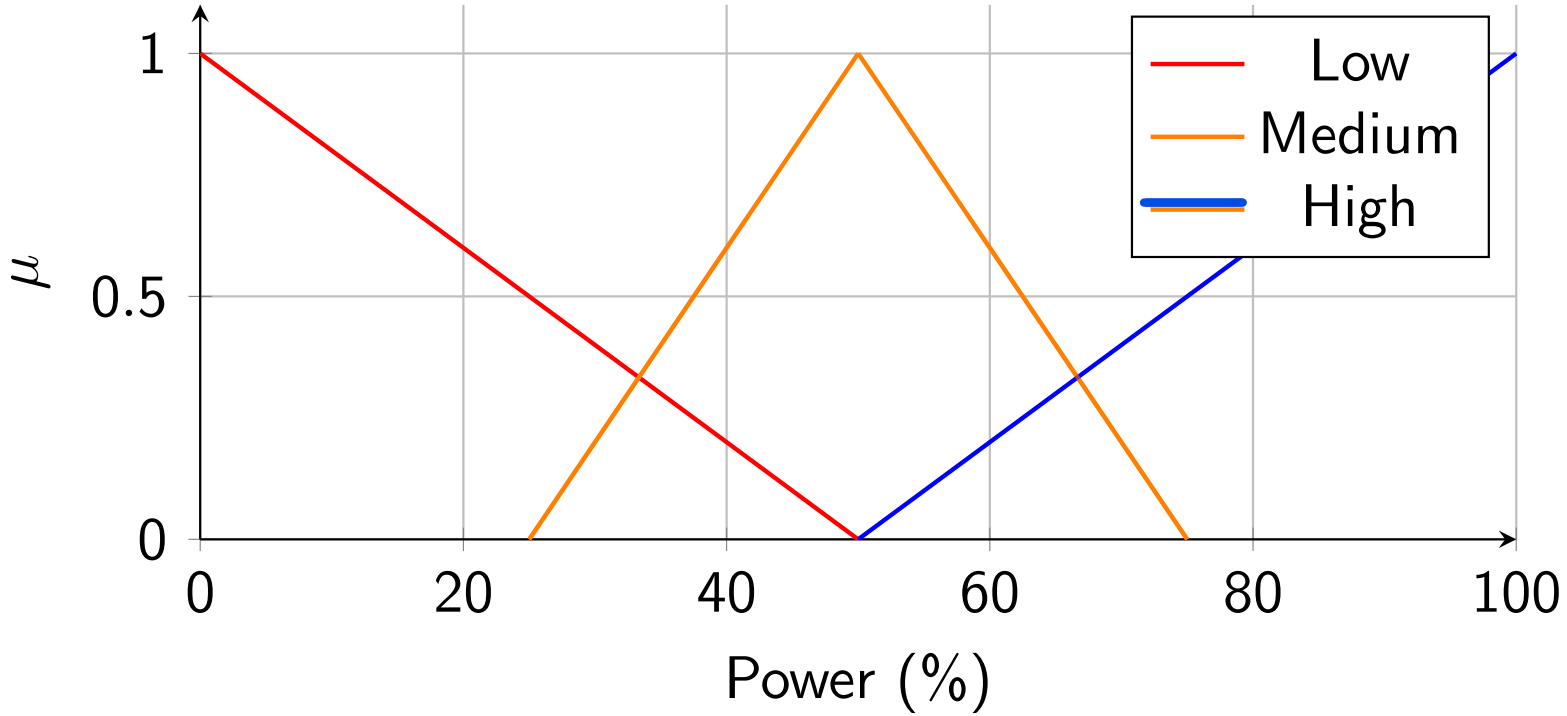
Mamdani Inference Rule Base

IF Temperature THEN Power

- IF Temperature is Cold THEN Power is High
- IF Temperature is Warm THEN Power is Medium
- IF Temperature is Hot THEN Power is Low

Mamdani Inference at 22°C

- Only "Warm" rule fires with strength 0.7
- Output fuzzy set: "Medium" Power clipped at 0.7



Defuzzification (Centroid)

- Use Centroid method:

$$x^* = \frac{\int x \cdot \mu(x)\, dx}{\int \mu(x)\, dx}$$

- Approximate centroid of the clipped "Medium" triangle
- Symmetrical triangle centered at 50 Centroid **50**
- So, Heater Power Output = **50%**

Final Output

- Input Temperature = 22°C
- Fuzzified: Warm = 0.7
- Rule Fired: "Warm $\rightarrow$ Medium Power"
- Clipped Output Set: Medium (0.7)
- Defuzzified Output: **50% power**

Heater should be set to 50% power.

Fuzzy Logic: Two-Input Fan Speed Controller

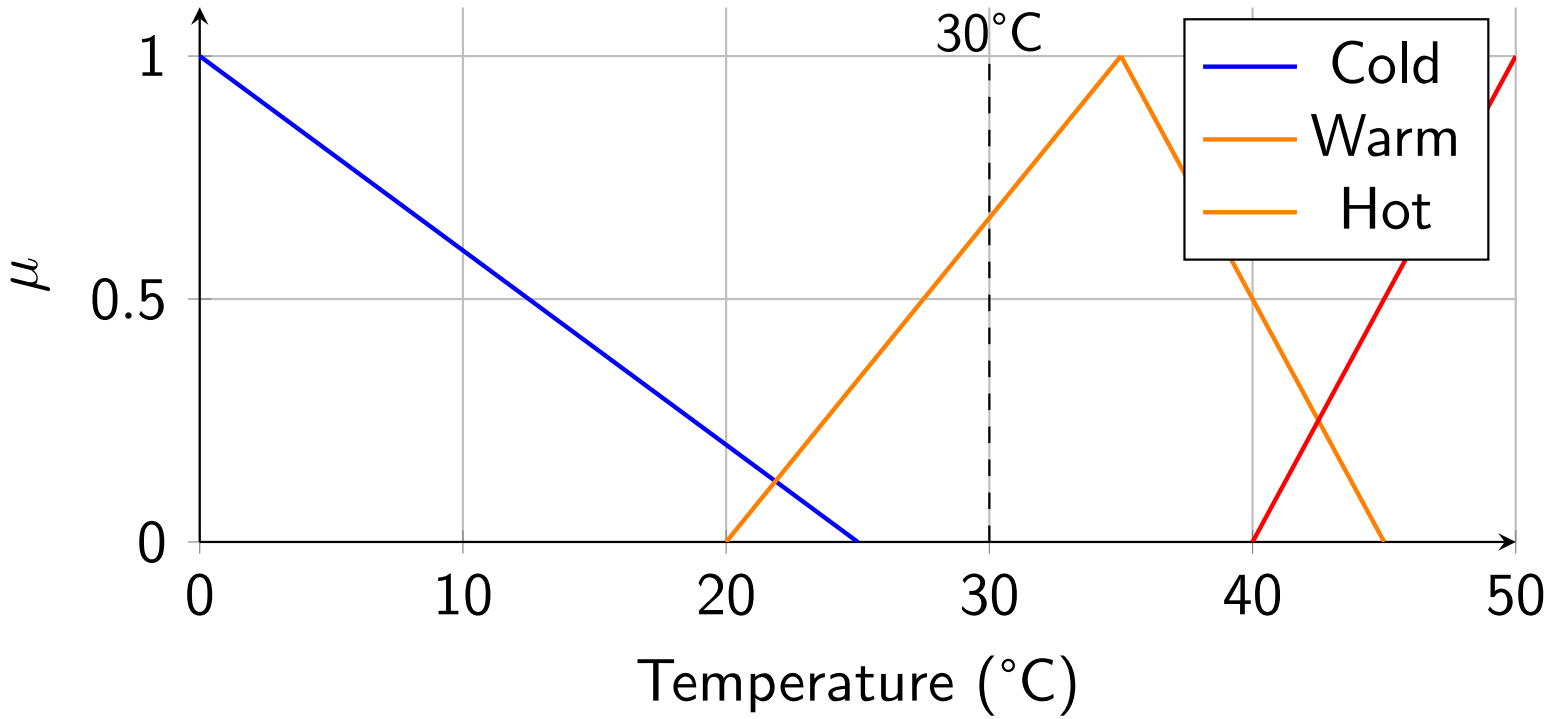
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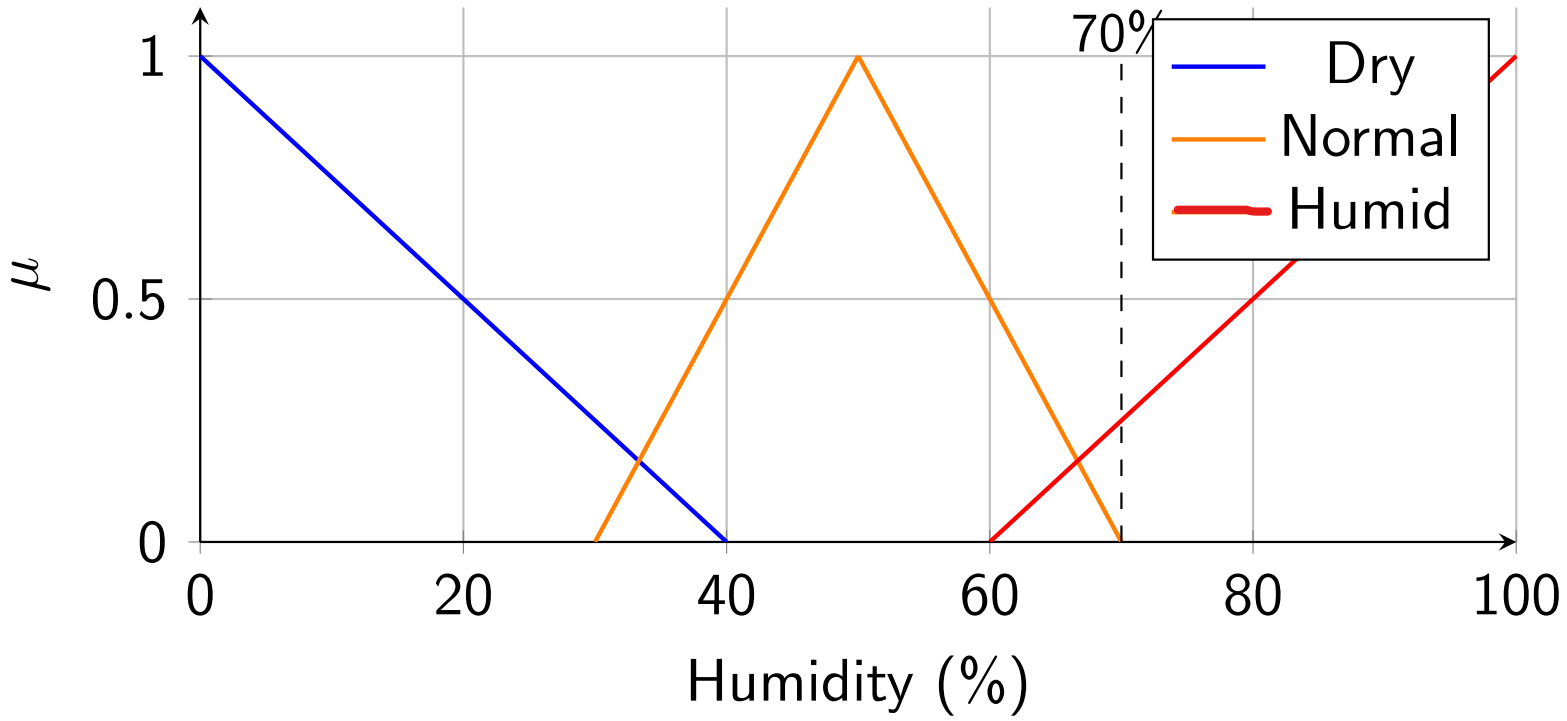
Problem Statement

- Inputs:
  - Temperature: Cold, Warm, Hot
  - Humidity: Dry, Normal, Humid
- Output: Fan Speed (Slow, Medium, Fast)
- Example: Temp = 30°C, Humidity = 70%

Fuzzification - Temperature



Fuzzification - Humidity



Mamdani Inference Rules

IF Temperature AND Humidity THEN Fan Speed

- IF Hot AND Humid THEN Fast
- IF Warm AND Normal THEN Medium
- IF Cold AND Dry THEN Slow
- IF Warm AND Humid THEN Fast

Active Rules for Input (30°C, 70%)

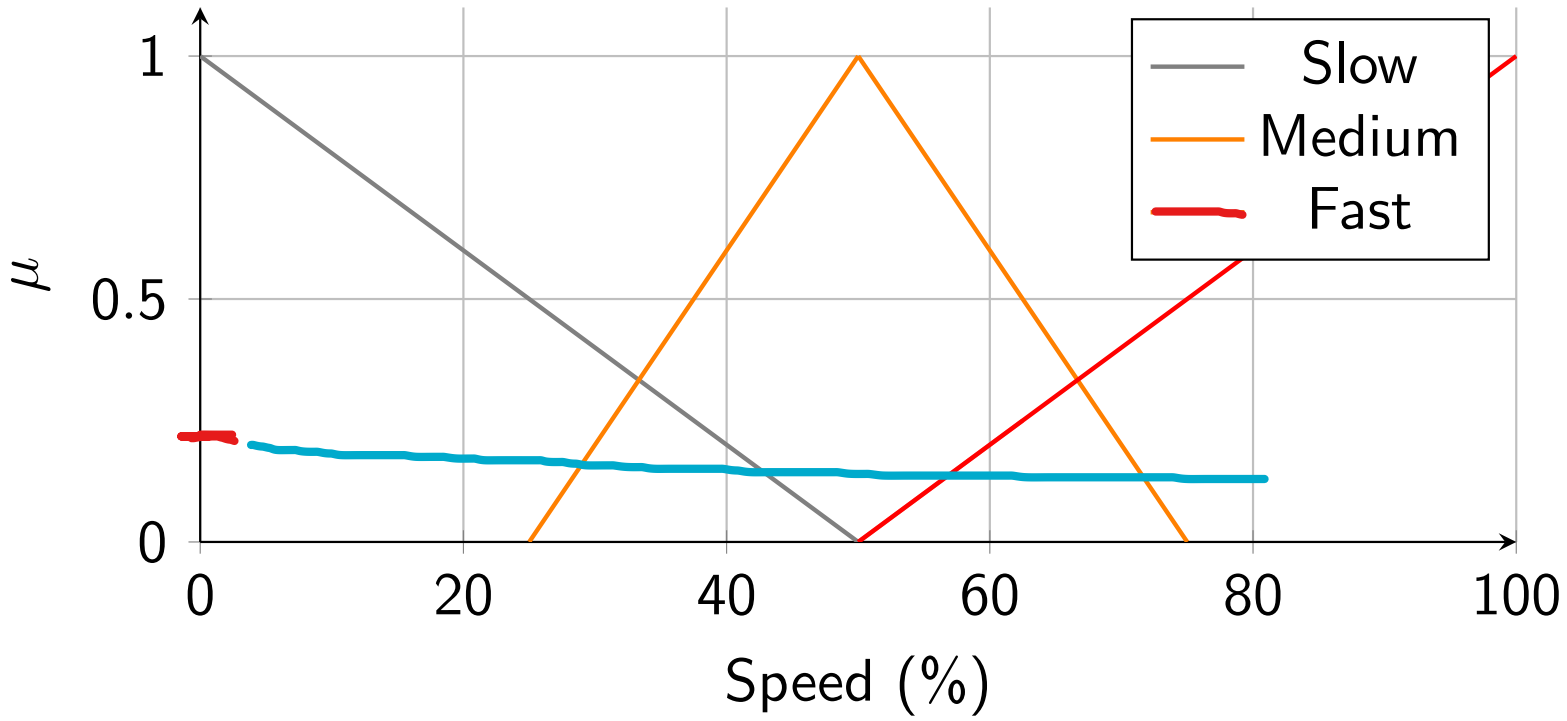
Temp = warm triangular,
x=30, a=20, b= 35, c=45

Humidity = humid triangular,
x=70, a=60(initial), b= 100(peak), c=?

- $\mu_{\text{Warm}}(30) = 0.67$ Here, a<x≤b
  so, x-a / b-a = 30-20 / 35-20 = 0.67
- $\mu_{\text{Humid}}(70) = 0.25$ Here, a<x≤b
  so, x-a / b-a = 70-60 / 100-60 = 0.25
- Active Rule: IF Warm AND Humid THEN Fast
- Firing strength $= \min(0.67, 0.25) = 0.25$

AND operator uses the minimum (min) function

Output Aggregation (Fan Speed)



Defuzzification (Centroid)

- Only one clipped triangle: Fast (at 0.25)
- Centroid 75% (peak of Fast is at 100, start at 50)
- Output: Fan Speed **75%** x= 50+100 / 2 = 75 %

Fuzzy Logic: Three-Input Ventilation Fan Controller

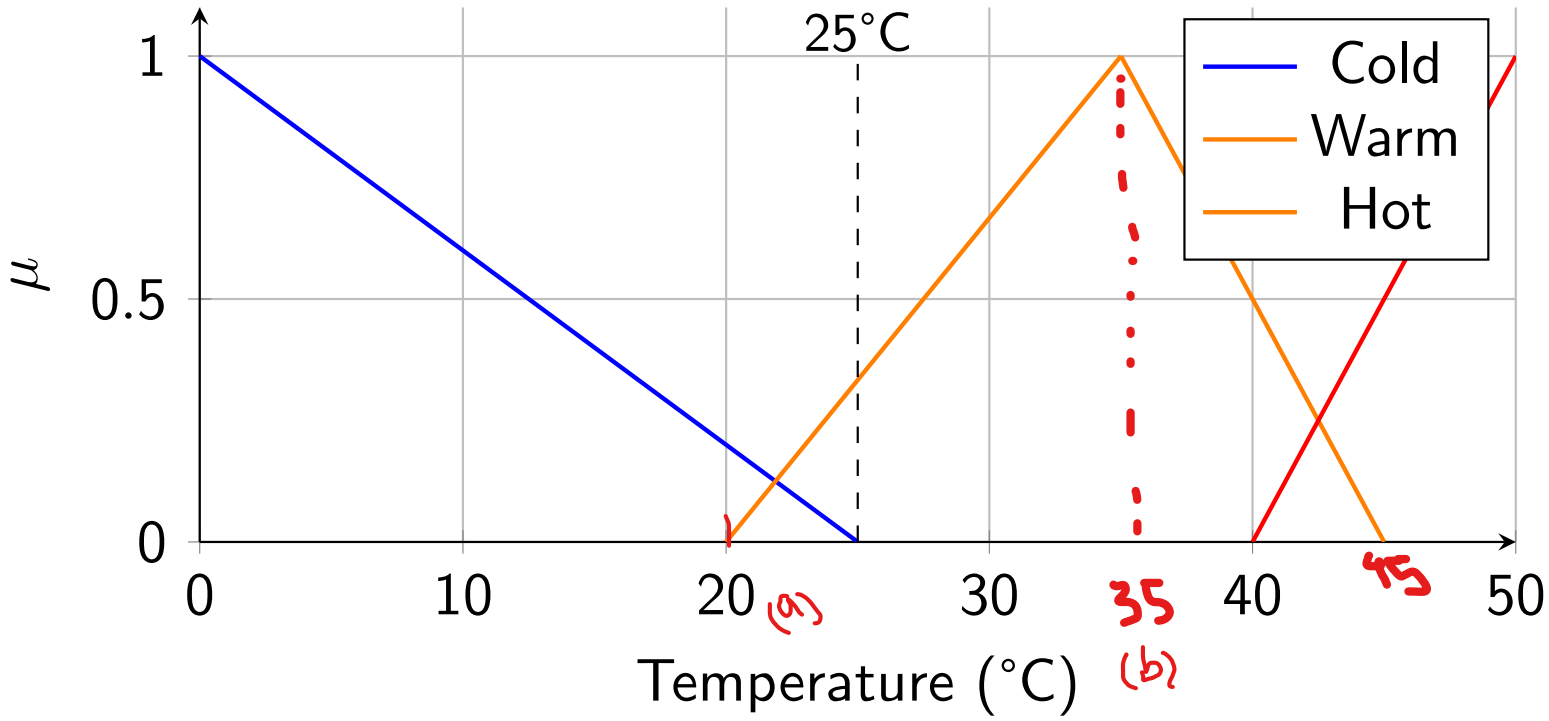
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Problem Statement

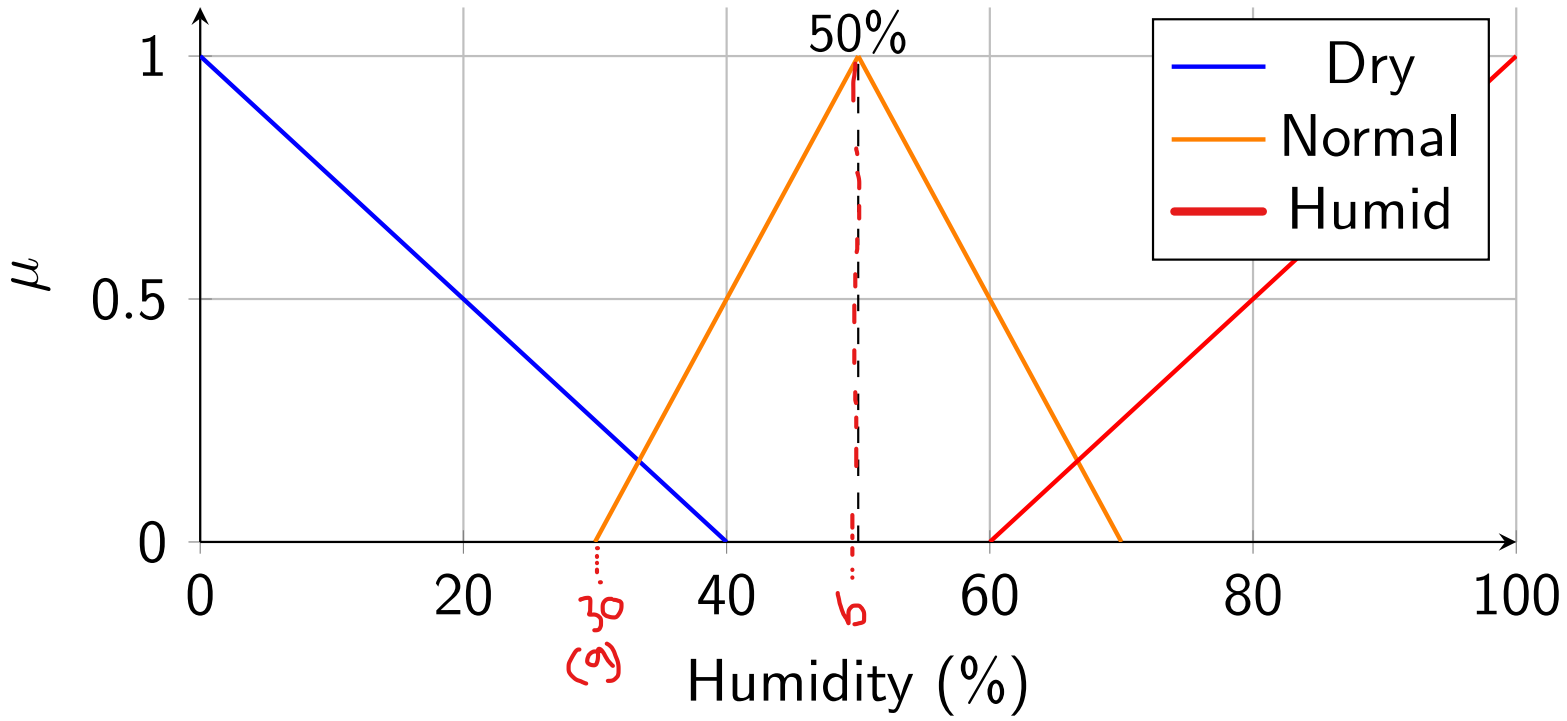
- Inputs:
  - Temperature: Cold, Warm, Hot
  - Humidity: Dry, Normal, Humid
  - CO Level: Low, Medium, High
- Output: Ventilation Fan Speed (Slow, Medium, Fast)
- Example: Temp = 25°C, Humidity = 50%, CO Level = 800 ppm

Fuzzification - Temperature



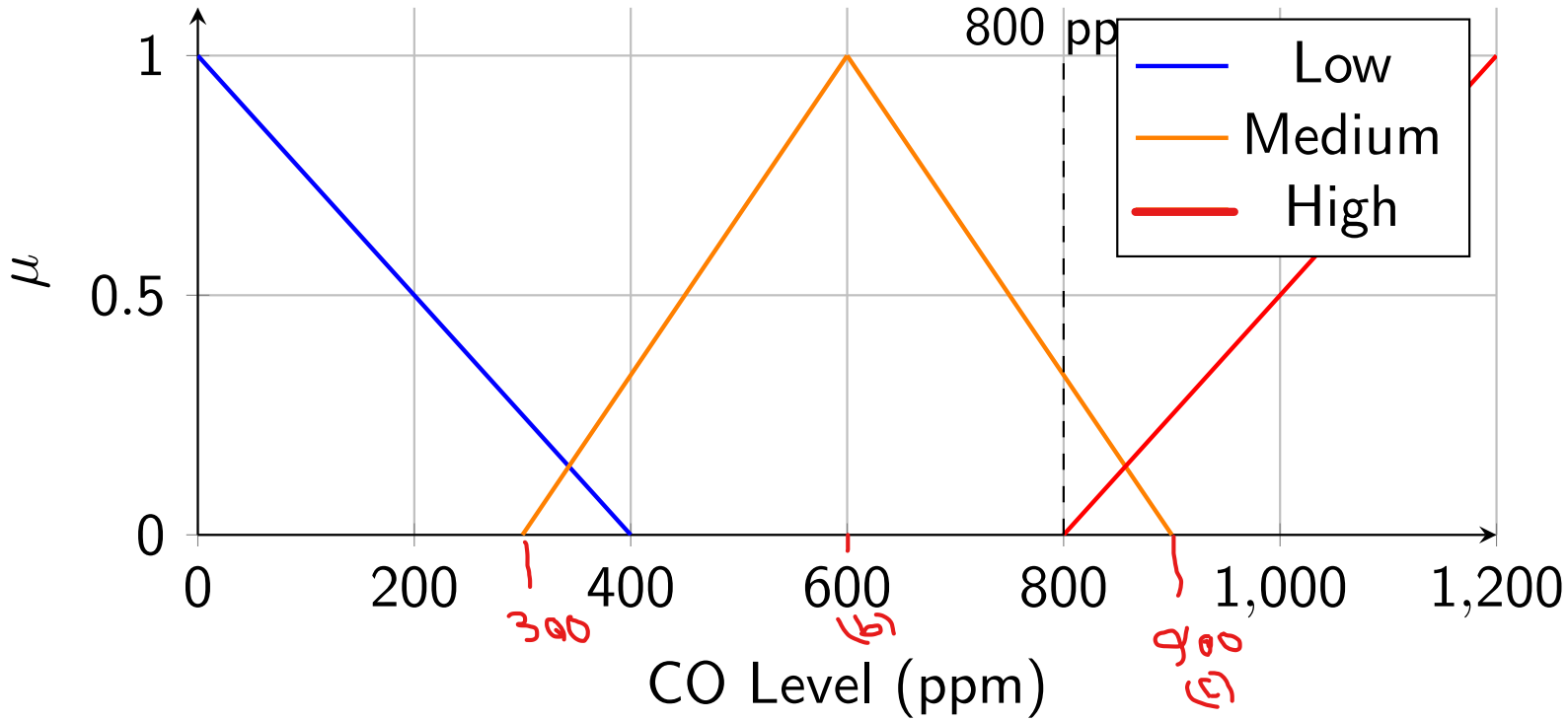
$$\frac{x-a}{b-a} = \frac{25-20}{35-20} = 0.33$$

Fuzzification - Humidity



$$\frac{x-a}{b-a} = \frac{50-30}{50-30} = 1$$

Fuzzification - CO Level



$$\frac{c-x}{c-b} = \frac{900-800}{900-600} = \frac{100}{300} = 0.33$$

Mamdani Inference Rules

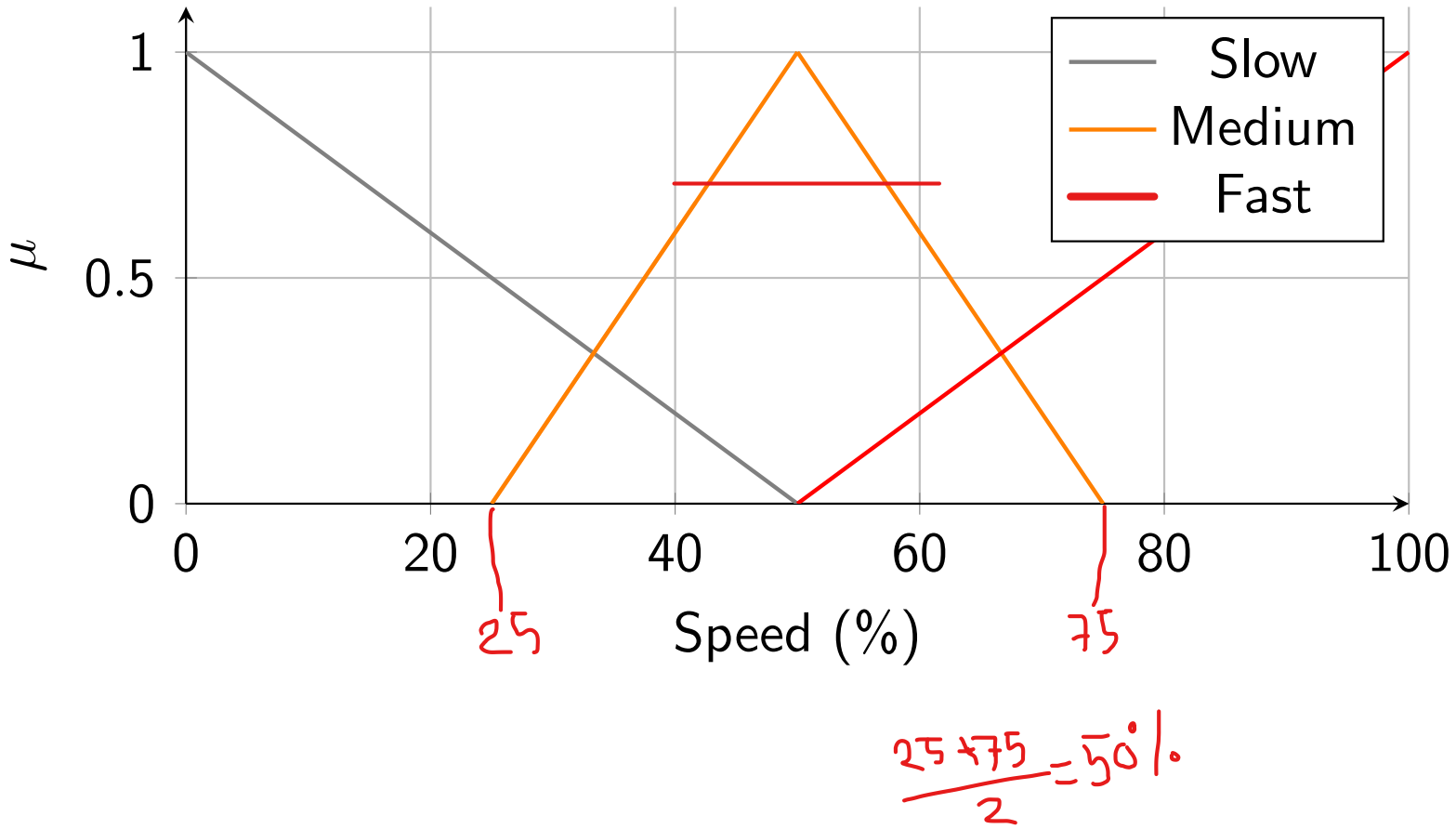
IF Temperature AND Humidity AND CO THEN Fan Speed

- IF Hot AND Humid AND High THEN Fast
- IF Warm AND Normal AND Medium THEN Medium
- IF Cold AND Dry AND Low THEN Slow
- IF Warm AND Humid AND High THEN Fast

Active Rules for Input (25°C, 50%, 800 ppm)

- $\mu_{\text{Warm}}(25) = 0.67$
- $\mu_{\text{Normal}}(50) = 0.75$
- $\mu_{\text{Medium}}(800) = 0.67$
- Active Rule: IF Warm AND Normal AND Medium THEN Medium
- Firing strength $= \min(0.67, 0.75, 0.67) = 0.67$

Output Aggregation (Fan Speed)



Defuzzification (Centroid)

- Centroid of the clipped Medium region $\approx$ 60% (Medium fan speed)
- Output: Fan Speed $\approx$ **60%**